# Calibration Behavior: 2024 vs. 2026

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## Overview

This report compares longitudinal calibration behavior between the 2024 and 2026 calibration datasets.

The analysis reproduces the drift analysis originally applied to the 2024 calibration data. For each anchor, calibration is compared with its calibration on the first day of the observation period.

Two measures of drift are examined:

1. **Instantaneous drift:** the absolute proportional difference between an anchor's current calibration and its calibration on Day 0.
2. **Maximum drift:** the largest instantaneous drift observed up to each point in time.

The analysis also examines the number of anchors that exceed 1%, 2%, 5%, 10%, and 20% deviation thresholds.

For the 2026 data, calendar dates are preserved throughout the analysis. Thus, the `days` variable represents actual elapsed calendar days rather than the number of calibration files retained after missing days were removed.

## 1. Load Data

The 2024 calibration data are read from individual CSV files. Each filename contains the date of retrieval, which is retained as the observation date, alongside the pre-computed calibration value.

The 2026 data requires us to calculate calibration value via given slope and intercept values.

The calibration value is calculated as:

$$y = 100m + b$$

where (m) is the slope and (b) is the intercept.

Table 1: Dataset Summaries

| year | anchors | observations | first_date | last_date | calendar_days |
|---|---|---|---|---|---|
| 2024 | 659 | 19111 | 2024-10-21 | 2024-11-18 | 28 |
| 2026 | 444 | 35520 | 2026-01-23 | 2026-04-14 | 81 |

Only anchors with complete observations across all retained calibration dates are used in the final comparison

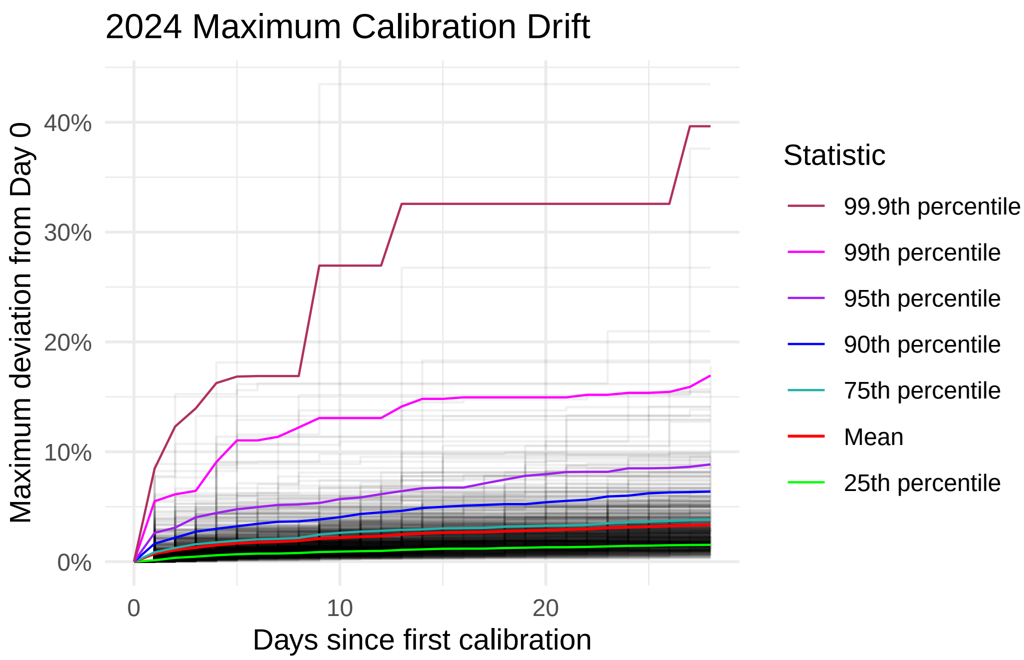
## 2. Calculating Drift

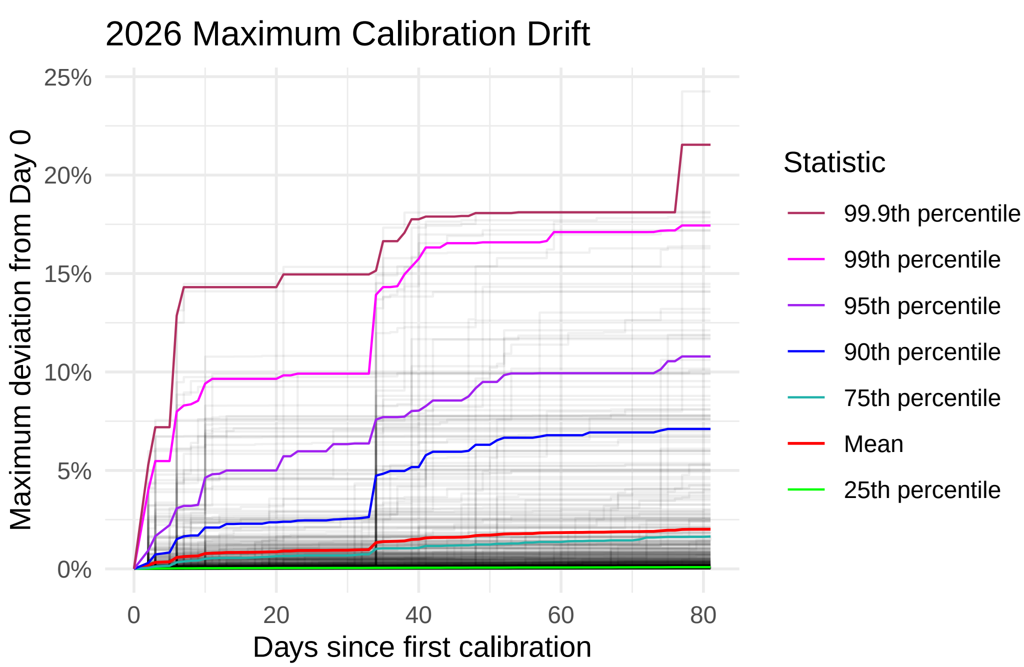
The following calculation is equivalent to Zach's original 2024 analysis.

For each anchor:

$$drift_t = \frac{|calibration_t - calibration_0|}{calibration_0}$$

## 3. Maximum Drift



### 3.1 90-99.9th Percentile Anchors

Which anchors were between the 90th and 99.9th percentile on at least once?

- 653 anchors were in the range at least once in 2024 data
- 444 anchors were in the range at least once in 2026 data
- 276 anchors were in the range at least once in both 2024 and 2026 data
- 185 anchors occupied the range only one time each in both 2024 and 2026 data
- 91 anchors occupied the range more than one time each between 2024 and 2026 data

### 3.2 Geographic Context of IDs Consistently in the 90-99.9 Range

Below, we take a look at the IDs that appeared in the 90-99.9th percentile range more than one time each between the 2024 and 2026 datasets. These anchors entered the range either once in 2024 and more than once in 2026, or more than once in 2024 and once in 2026, or entered the range more than once in both 2024 and 2026 data. We will discuss the geographic context of these 91 anchors below:

Table 2: ASN Frequency

| asn | anchors |
|---:|---:|
| 14061 | 4 |
| 16276 | 4 |
| 3491 | 3 |
| 57695 | 3 |
| 36236 | 2 |
| 197540 | 2 |

Table 3: Continent Frequency

| continent | anchors |
|---|---:|
| Europe | 59 |
| North America | 15 |
| Asia | 14 |
| Oceania | 3 |

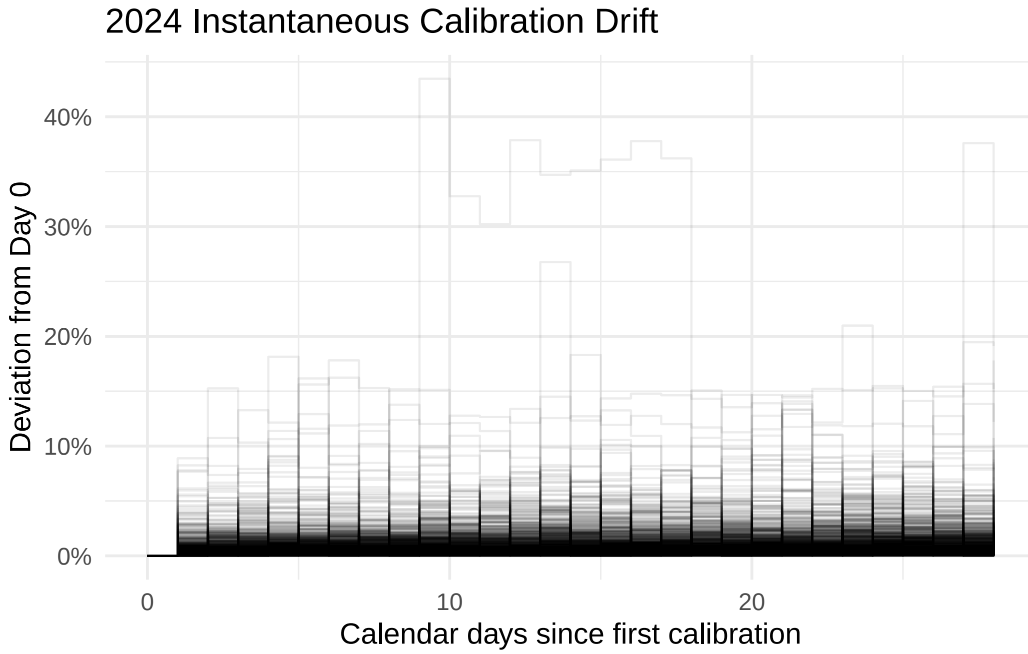
Table 4: City Frequency

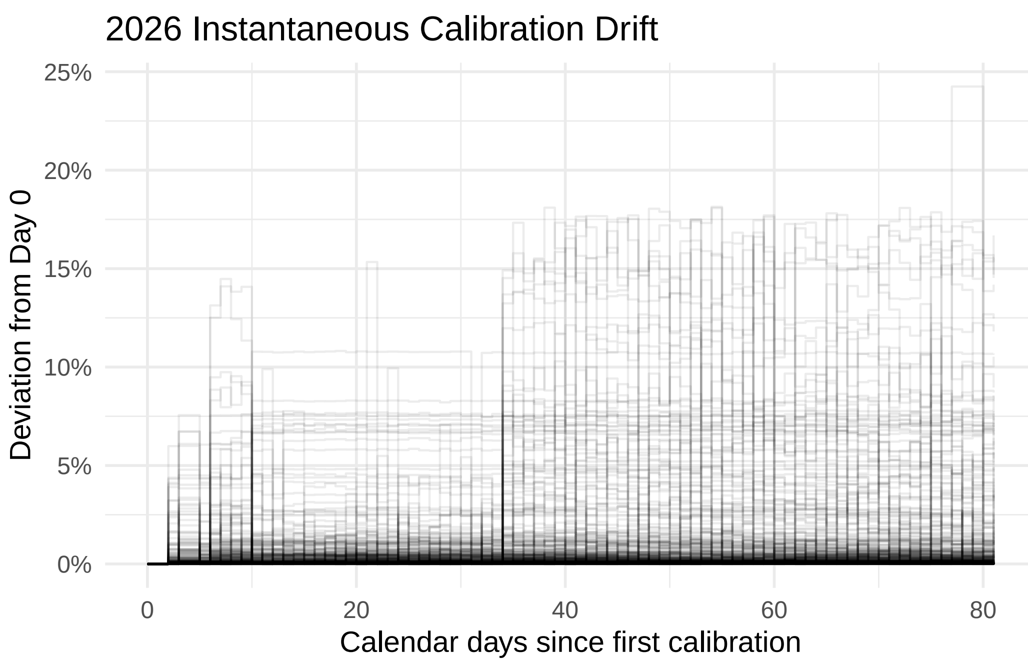
| city | anchors |
|---|---|
| Singapore | 8 |
| Frankfurt | 4 |
| Prague | 4 |
| Seattle | 4 |
| Vienna | 4 |
| Frankfurt am Main | 3 |
| Amsterdam | 2 |
| Berlin | 2 |
| Dubai | 2 |
| Hong Kong | 2 |
| London | 2 |
| Nuremberg | 2 |
| Phoenix | 2 |
| Sydney | 2 |

Table 5: Country Frequency

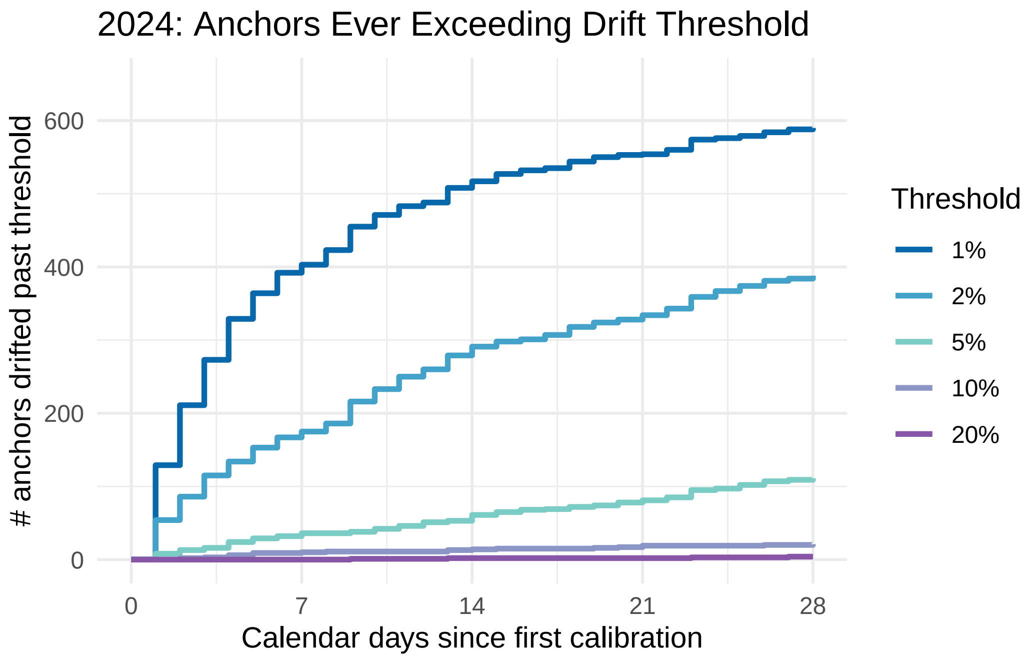
| country | anchors |
|---|---|
| DE | 16 |
| US | 14 |
| SG | 8 |
| AT | 7 |
| FR | 6 |
| IT | 6 |
| CZ | 5 |
| GB | 5 |
| NL | 3 |
| AE | 2 |
| AU | 2 |
| HK | 2 |
| IN | 2 |
| PL | 2 |
| SE | 2 |

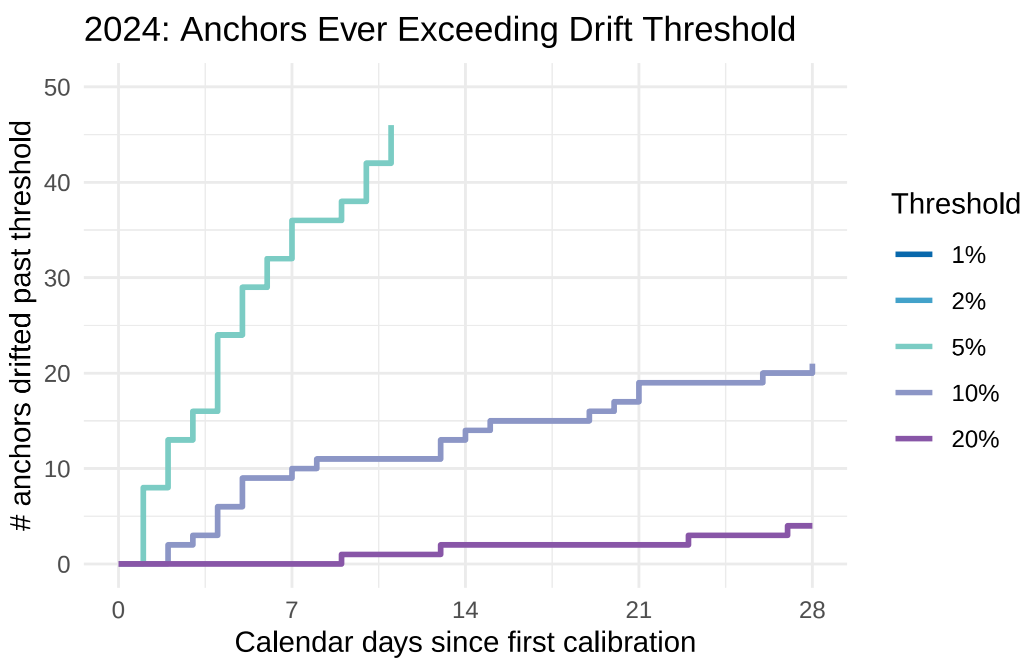
## 4. Instantaneous Drift

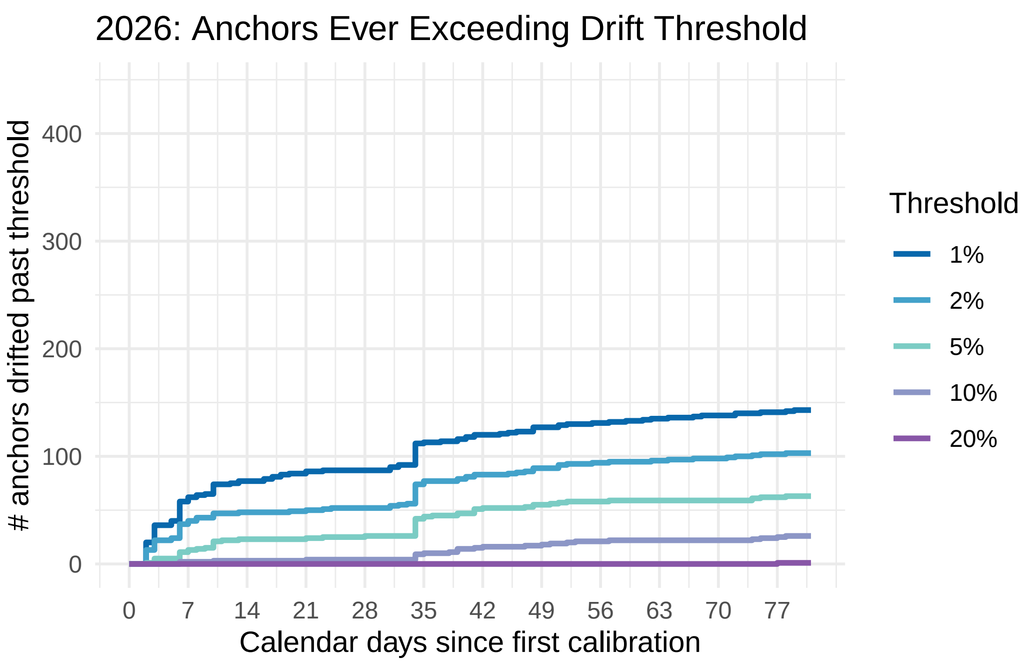


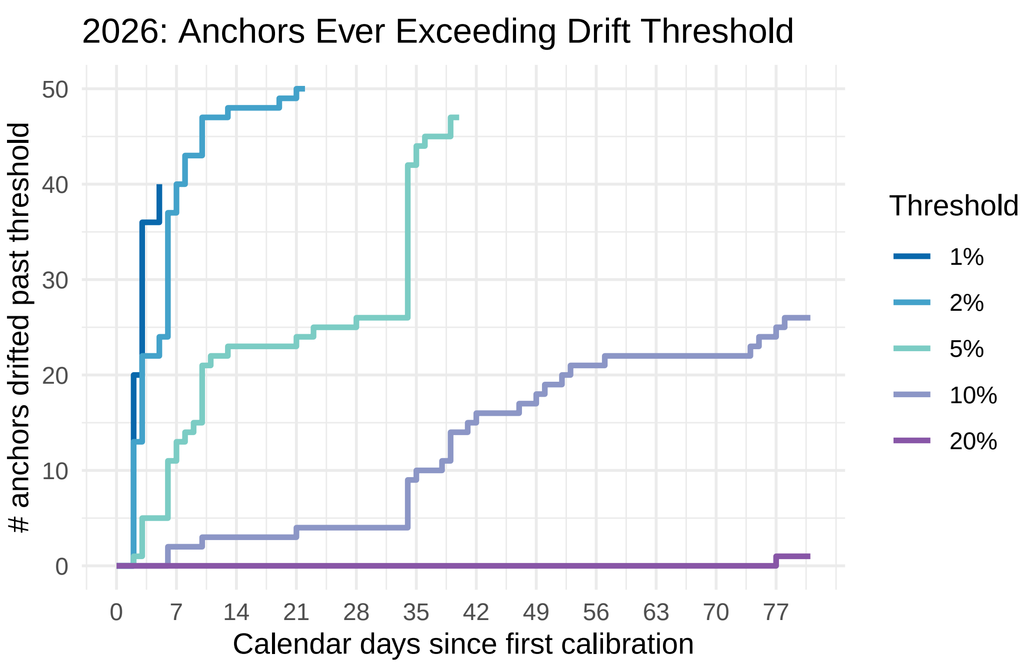



## 5. Maximum Drift Threshold Analysis



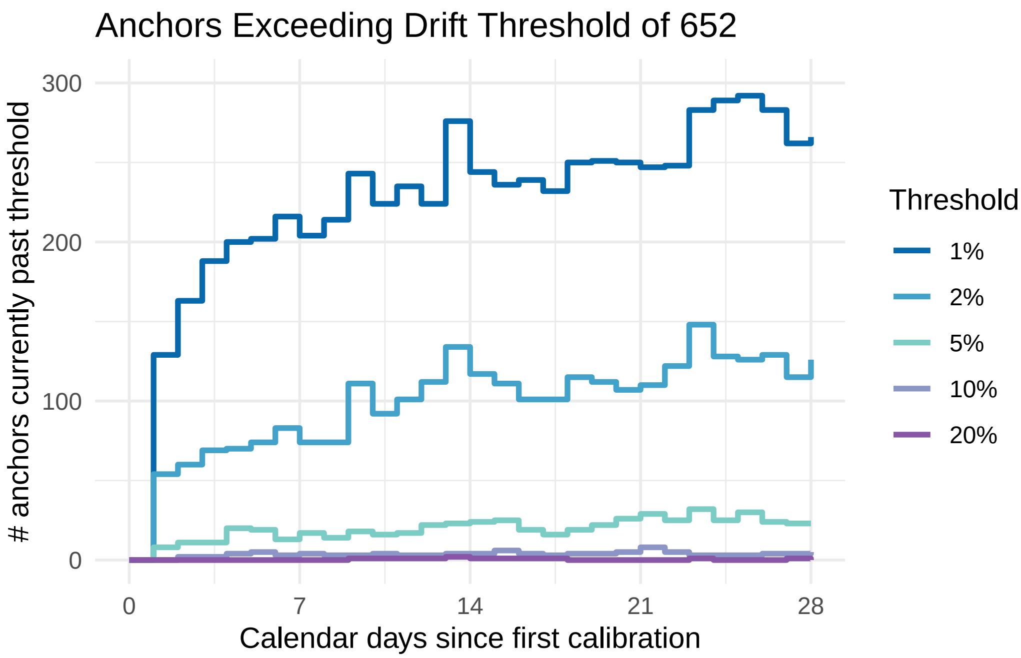



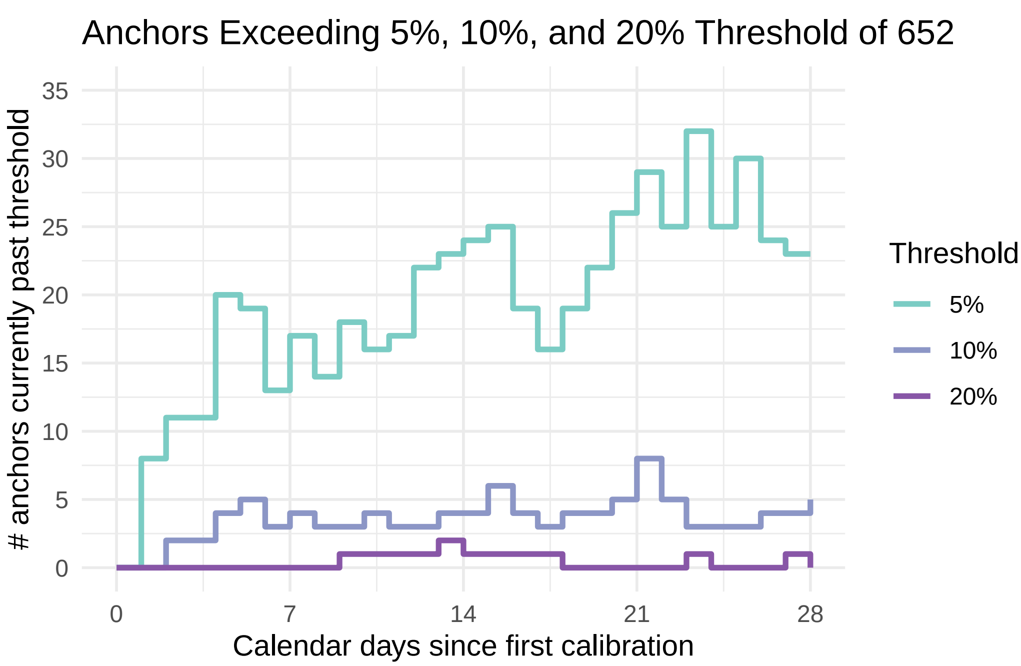

2026: Anchors Ever Exceeding Drift Threshold
# anchors drifted past threshold
400
300
200
100
0
0
7
14
21
28
35
42
49
56
63
70
77
Calendar days since first calibration
Threshold
1%
2%
5%
10%
20%

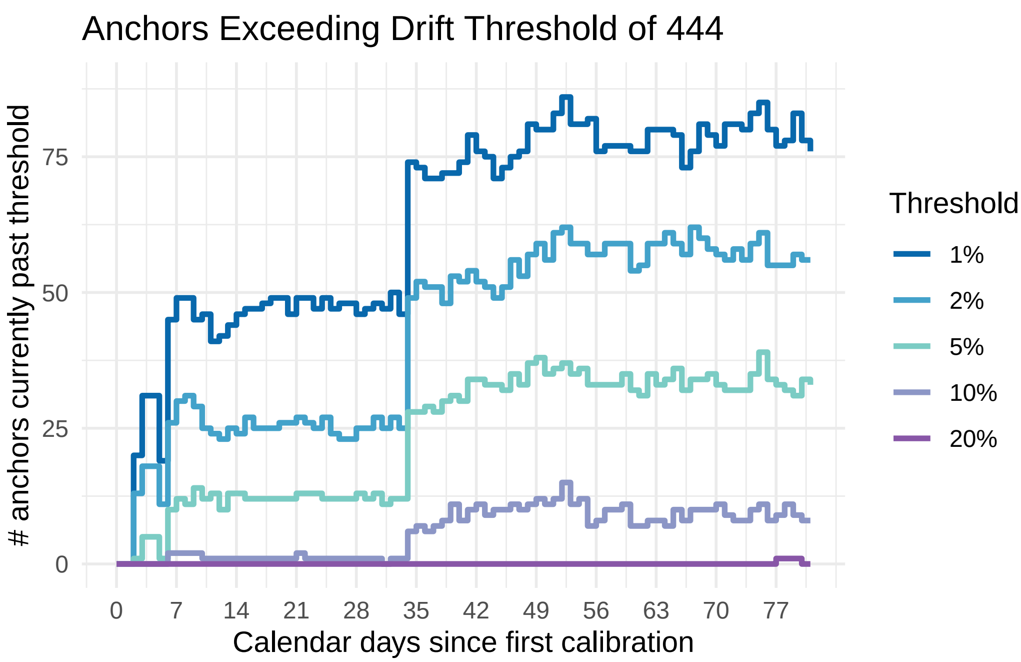

2026: Anchors Ever Exceeding Drift Threshold
# anchors drifted past threshold
50
40
30
20
10
0
0
7
14
21
28
35
42
49
56
63
70
77
Calendar days since first calibration
Threshold
1%
2%
5%
10%
20%

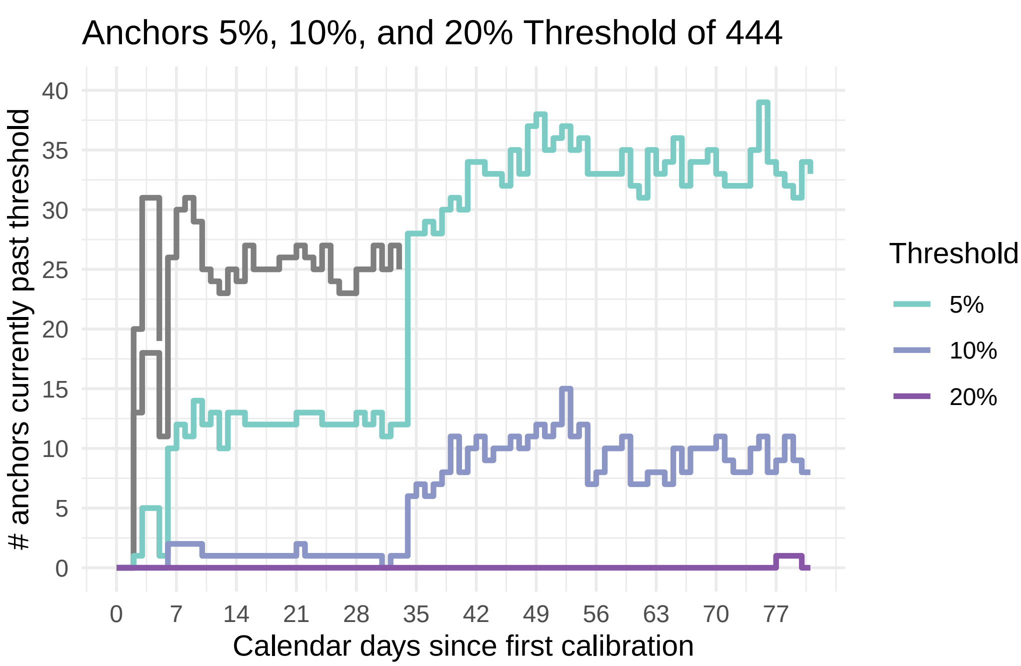
## 6. Instantaneous Drift Threshold Analysis

This analysis counts the number of anchors that are currently beyond each threshold, rather than the number that have ever exceeded it.



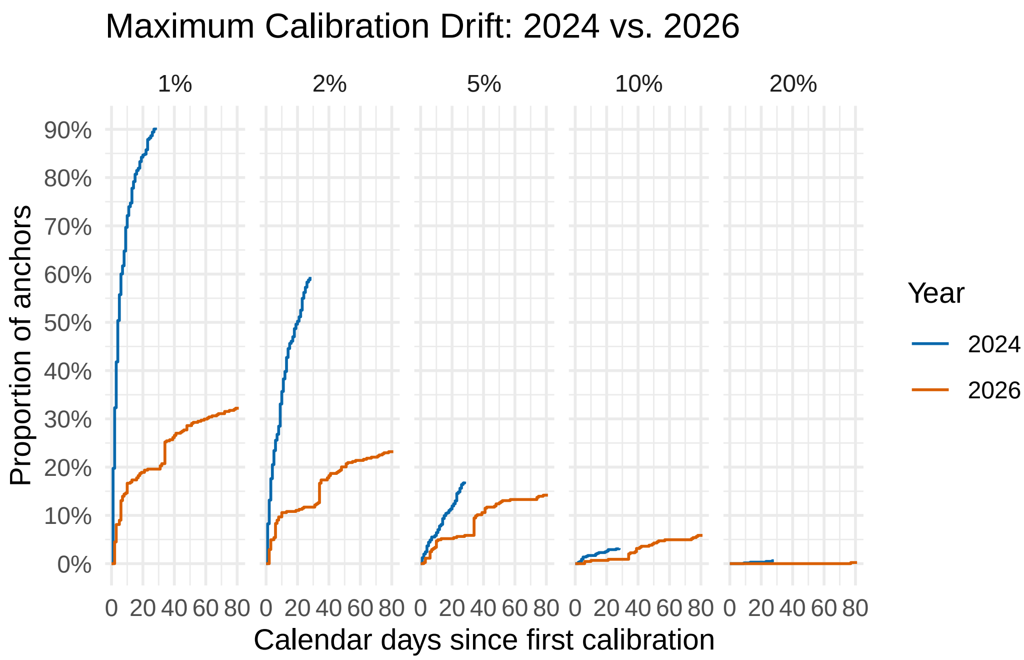



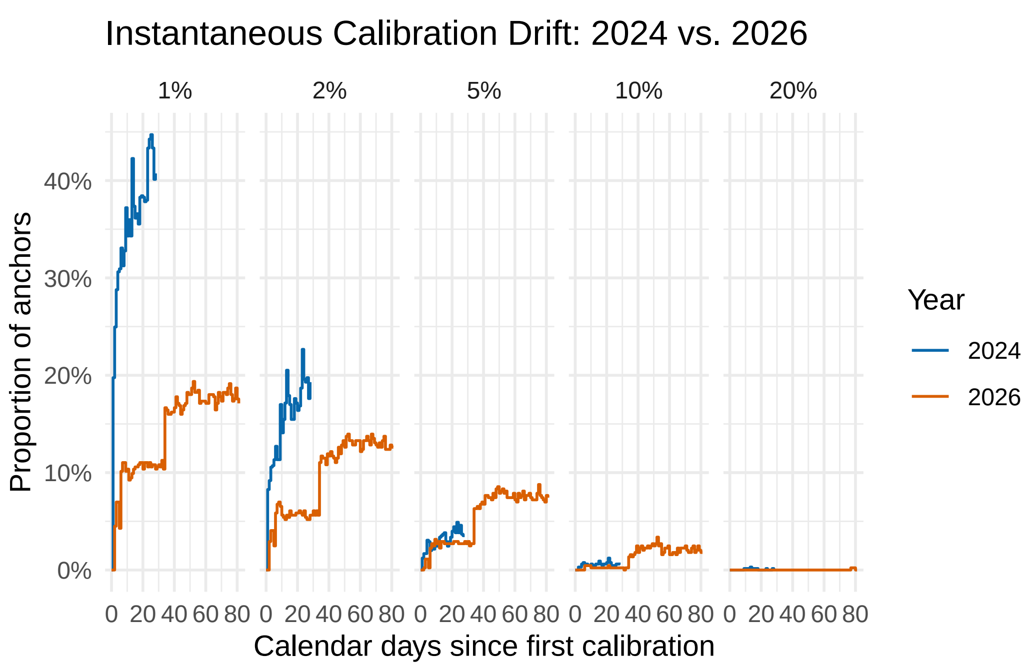

Anchors Exceeding Drift Threshold of 444
# anchors currently past threshold
Calendar days since first calibration
0
7
14
21
28
35
42
49
56
63
70
77
0
25
50
75
Threshold
1%
2%
5%
10%
20%


Anchors 5%, 10%, and 20% Threshold of 444
# anchors currently past threshold
Calendar days since first calibration
0
7
14
21
28
35
42
49
56
63
70
77
0
5
10
15
20
25
30
35
40
Threshold
5%
10%
20%

## 7. Direct 2024 vs. 2026 Comparison

For direct comparison, it is useful to put both years onto the same plot. Because the datasets contain different numbers of anchors, percentages are more informative than raw counts.



## 8. Notes for Interpretation

The maximum-drift plots measure whether an anchor has **ever** departed from its initial calibration by a given amount. Consequently, these curves can only increase or remain constant.

The instantaneous-drift plots measure how many anchors are **currently** beyond each threshold. These curves can increase or decrease as anchors move back toward their initial calibration.

A pattern in which maximum drift increases over time while instantaneous drift remains approximately stable would suggest that individual anchors experience deviations but that the overall population does not undergo a systematic accumulation of drift.

The 2024 and 2026 comparisons should be interpreted using calendar time. Missing calibration dates in the 2026 dataset remain gaps in the time axis rather than being compressed into consecutive observation numbers.

The **last two plots** in this QMD are probably the most useful for your actual comparison. They put 2024 and 2026 on the **same percentage scale**, so the fact that you have 444 complete anchors in 2026 versus a different number in 2024 doesn't distort the comparison.